

5

patterns & relationships

One Thing, Another Thing

when one number predicts another

Yesterday's numbers all stood on their own. *How many subscribers? How long is a typical session? Total hours watched?* Useful — but every one of them is about **one thing in isolation**.

Today we look at **relationships** — questions like *does longer tenure mean more watch hours?* The tool for spotting them is the scatter plot, plus a line of best fit and a trust-score called R^2 .

By the end of today: scatter charts, one-click trendlines, the equation, and a feel for when to trust it.

Is this where we hit proper maths?

— you, eyeing the page suspiciously

It looks like maths. It's actually just three numbers and a line.

the three numbers: slope, intercept, R^2 .



BRAIN POWER

Before any maths: do you think a longer-tenure subscriber watches more, less, or about the same as a newer one? Jot your guess down. We'll check what the data actually says on page 10.

turn over — we draw a picture first →

What predicts what?

Same desk, same coffee, slightly different question. Yesterday: *what's the average?*

Today: *do these two things move together?*

RELATIONSHIPS

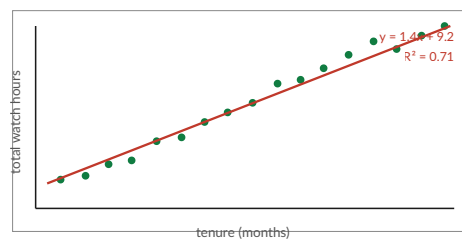
tenure vs hours?
rating vs views?
plan vs duration?

draw the picture
check the trust score

Priya:

"Do loyal customers
actually watch more,
or do we just
assume they do?"

Excel — tenure vs hours



DAY 5

Plot tenure vs watch hours

A scatter plot needs exactly two numeric columns — one for X, one for Y. Today's question is *"do longer-tenure subscribers watch more?"*, so:

- **X-axis (horizontal):** `tenure_months` — how long each subscriber has been with StreamWave.
- **Y-axis (vertical):** `total_watch_hours` — how many hours each one has watched.

Neither column exists yet. Build them first.

Step 1 — Add the two columns to your Subscribers sheet

On the Subscribers sheet (loaded from Power Query yesterday), find the first empty column to the right of the Table. Excel will offer to extend the Table — let it.

New column 1: name it `tenure_months`. Formula:

```
=DATEDIF([@join_date], TODAY(), "M")
```

DATEDIF returns the whole months between two dates. `[@join_date]` is the structured reference to the current row's `join_date`.

New column 2: name it `total_watch_hours`. Formula:

```
=SUMIF(tblActivity[subscriber_id], [@subscriber_id], tblActivity[duration]) / 60
```

SUMIF totals all the duration values from the Activity table where the `subscriber_id` matches the current row's ID. Divide by 60 to convert minutes to hours.

Step 2 — Select the two columns and insert the chart

Click the `tenure_months` header, then Ctrl-click the `total_watch_hours` header so both columns are selected. **Insert** → **Charts** → **Scatter** → pick the first option (dots only, no lines). Excel drops a scatter plot onto the sheet.

Step 3 — Eye-test before maths

Before doing anything else, look at the shape:

- **Climbing up-and-right** → positive relationship.
- **Falling down-and-right** → negative.
- **Blob** → no relationship.
- **Curve** → a relationship, but not a straight one.

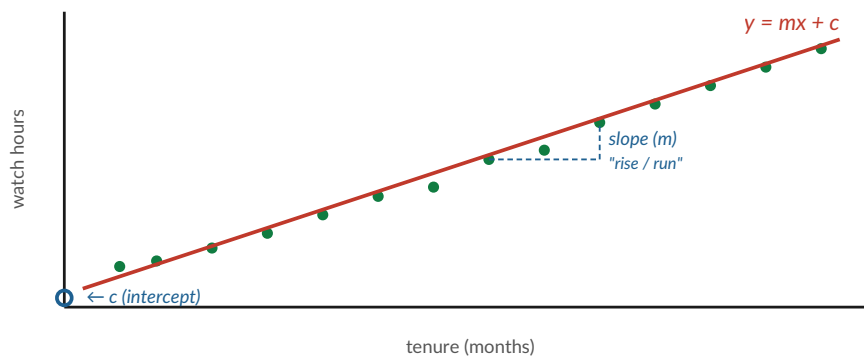
$y = mx + c$

Once your scatter plot shows a roughly straight-line pattern, the line of best fit is a single line drawn through the dots that minimises the total distance to all of them. It summarises the pattern in **two numbers**: a slope (m) and an intercept (c).

In plain English

- **Slope (m)** = how much Y changes when X goes up by 1. If $m = 1.4$ hours per month, then every extra month of tenure adds about 1.4 watch hours.
- **Intercept (c)** = where the line crosses the Y-axis. The "starting value" when $X = 0$. If $c = 9.2$ hours, a brand-new subscriber (zero tenure) is predicted to watch about 9.2 hours.

Together, the line is $y = mx + c$. For our example: $\text{hours} = 1.4 \times \text{tenure} + 9.2$. Plug any tenure value in, get a predicted hours value out. That's all the maths is doing.



A line drawn through a scattering of dots *summarises* the relationship in two numbers. The dots don't all sit on the line, but the line is the best straight-line story we can tell about them.

Add a trendline

You don't draw the line of best fit yourself. Excel does it. Six clicks, and the equation and R^2 appear right on the chart.

The procedure

1. Click anywhere on the chart you just inserted, so it's selected (a thin border appears around it).
2. **Right-click any dot** in the plot area. Make sure the cursor is on a dot, not the white space between dots — if you right-click white space you get a different menu.
3. From the pop-up menu, click **Add Trendline...** A pane called *Format Trendline* opens on the right of the screen.
4. In that pane, under *Trendline Options*, **Linear** is already selected. Leave it.
5. Scroll down to the bottom of the same pane. You'll see two checkboxes.
6. Tick "**Display Equation on chart**".
7. Tick "**Display R-squared value on chart**".
8. Close the pane (the X at its top-right corner). The equation and R^2 now sit on the chart, usually near the line.

[SCREENSHOT 1 — FORMAT TRENDLINE PANE]

The Format Trendline task pane open, with the Trendline Options section visible. Linear radio button selected. Both checkboxes at the bottom ticked. Chart in the background showing dots, a red trendline, the equation and R^2 visible in the plot area.

What you should now see on the chart

$$y = 1.4x + 9.2$$

$$R^2 = 0.71$$

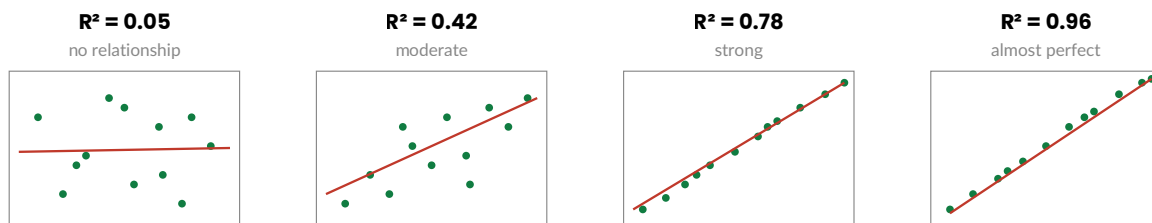
(Your numbers will be similar but not identical — depends on exactly which subscribers' data you're using.) The equation gives you slope (1.4) and intercept (9.2). The R^2 gives you the trust score. The next page tells you what to do with each.

R^2 — the trust score

The trendline is a *summary* of a relationship. R^2 tells you how good that summary is. It's a number between 0 and 1: how much of the variation in Y is explained by X.

The scale, in plain English

- $R^2 = 1.0$ — every dot sits exactly on the line. Y is perfectly predicted by X. *Almost never happens with real data.*
- $R^2 \approx 0.7+$ — strong relationship. The line is a good summary; X explains most of what Y is doing.
- $R^2 \approx 0.4$ to 0.7 — moderate relationship. There's something there, but X is only part of the story. Other factors matter too.
- $R^2 \approx 0.1$ to 0.4 — weak relationship. The line tilts but the dots are scattered all over. Don't lean on the line.
- $R^2 \approx 0$ — no relationship. Don't quote the equation. The dots are a blob.



ASSESSOR FAVOURITE

If asked "what does an R^2 of 0.71 mean?", the one-sentence answer: "It means about 71% of the variation in Y is explained by X. The other 29% is down to other factors not in this analysis." Strong, plain English, no jargon.

SLOPE, INTERCEPT, RSQ

The trendline puts the three numbers on the chart. Sometimes you want them as values in a summary table — for example, to compare slopes across regions or plans. Three Excel formulas do exactly that.

Function	Returns	Syntax
SLOPE	The slope (m)	<code>=SLOPE(known_y, known_x)</code>
INTERCEPT	The intercept (c)	<code>=INTERCEPT(known_y, known_x)</code>
RSQ	R^2 — the trust score	<code>=RSQ(known_y, known_x)</code>

Be careful with argument order

All three functions take **Y first, then X**. That's the opposite of how you read the equation $y = mx + c$. It's the number-one mistake people make. To compute the relationship between tenure and watch hours, where hours is Y and tenure is X:

```
=SLOPE(tblSummary[watch_hours], tblSummary[tenure_months])
```

A useful bonus — CORREL

Closely related: `=CORREL(range1, range2)` returns the Pearson correlation coefficient — a number from -1 to +1 that measures the *direction and strength* of the linear relationship. CORREL squared equals RSQ. CORREL keeps the sign (negative for "as X goes up, Y goes down"), R^2 doesn't.

SHOW THIS OFF

When building a summary table, prefer the formulas to the chart numbers. *"I used =SLOPE, =INTERCEPT and =RSQ so the three numbers live in cells I can reference. The chart is for the audience; the cells are for the methodology."*

Halfway House

Four R^2 values, four short scenarios. Match the right description to each. Answers below the line; wrong answer = the page you should re-read.

1. You plot session duration vs avg_user_rating. The dots are everywhere. $R^2 = 0.03$.

☐ **Strong** ☐ **Moderate** ☐ **Weak** ☐ **No relationship**

2. You plot tenure_months vs total_watch_hours per subscriber. Dots climb up-and-right. $R^2 = 0.71$.

☐ **Strong** ☐ **Moderate** ☐ **Weak** ☐ **No relationship**

3. You plot release_year vs avg_user_rating. The dots drift down-and-right gently. $R^2 = 0.31$.

☐ **Strong** ☐ **Moderate** ☐ **Weak** ☐ **No relationship**

4. You plot watch_date (day number) vs cumulative sessions over the year. Almost a perfect line. $R^2 = 0.98$.

☐ **Strong** ☐ **Moderate** ☐ **Weak** ☐ **No relationship**

Answers — check yourself

1. **No relationship.** R^2 near 0 means session duration tells you essentially nothing about how viewers rated the title.

2. **Strong.** $R^2 = 0.71$ means about 71% of variation in hours is explained by tenure. Worth reporting confidently.

3. **Weak.** 31% is something but nearly 70% of the variation comes from elsewhere. Mention as a soft pattern, don't lean on it.

4. **Almost perfect.** A cumulative-over-time line is almost always strongly linear. R^2 this high should make you ask "is this a real finding or just a property of cumulative counting?" — often the latter.

Three pitfalls

Regression looks like maths so people trust it more than they should. Three things can wreck your conclusion even when the R^2 looks good.

1. Correlation isn't causation

Two things moving together doesn't mean one causes the other. Subscribers who watch a lot of Drama might also rate Drama highly — but watching more doesn't *cause* them to rate higher. A third factor (they like Drama in the first place) drives both. Always ask: "*could there be a lurking variable here?*" If you can think of one, name it. Don't claim causation from a scatter plot.

2. Tiny samples lie

An R^2 of 0.95 from four data points is meaningless. A line will always fit a small number of dots well — that's just geometry, not insight. As a rough rule: at least 20–30 paired observations before a high R^2 means anything. If you have fewer, report the chart and the equation but be loud about the sample size.

3. Cherry-picked ranges

The relationship between tenure and hours might be strong over the first 12 months and then flatten out. If you only plot the first 12 months, you'll get a great R^2 and a confident slope — and you'll be wrong about everyone over a year tenure. Always plot the *full* range your data covers, even if it's messier.

ASSESSOR FAVOURITE

Whenever you present a regression finding, end the sentence with a *caveat*. "*Tenure explains 71% of watch hours variation — but there's likely a lurking variable around plan type that I haven't accounted for.*" The caveat *builds* credibility. It shows you understand the limits of the analysis.

Verify the regression — sense-check before you trust it

Yesterday's named device — **The Two-Route Check** — applies to regression too. Before you quote slope, intercept, or R^2 to Priya, run three quick verifications:

1. **Eyeball the scatter plot.** Does the trendline actually look like it fits the cloud of dots? A high R^2 with an obvious outlier pulling the line is worse than no line at all. If the line goes through the dots, you're fine. If it sits above or below most of them, something's wrong.

2. **Predict and compare.** Pick a real subscriber. Tenure = 24 months. The formula says watch hours = $1.4 \times 24 + 9.2 = 42.8$. Look up that subscriber's actual watch hours. They should be near 42.8 – within 30% or so. If they're a long way off, the line doesn't generalise. (This is the Two-Route Check for regression: model prediction vs actual.)
3. **Compute SLOPE/INTERCEPT/RSQ two different ways.** Once from the chart trendline, once from the formulas. The numbers should match to two decimal places. If they don't, your data range is wrong somewhere.

None of these takes more than 30 seconds. All three together is what makes the difference between "I drew a trendline" and "I drew a trendline and verified it." Assessors notice the latter.

Interpret your result

You built the columns on page 3. You added the trendline on page 5. Your chart now looks something like this:

$$y = 1.4x + 9.2 \quad R^2 = 0.71$$

Read it as three statements

- **Slope (1.4).** Every extra month of tenure is associated with about 1.4 more total watch hours.
- **Intercept (9.2).** A brand-new subscriber (zero tenure) is predicted to watch about 9.2 hours.
- **R² (0.71).** About 71% of the variation in watch hours is explained by tenure. The other 29% is down to other factors not in this analysis.

Write the finding for Priya

Three numbers, one sentence of plain English, one caveat. Put this in the supporting notes for the dashboard:

"Longer-tenure subscribers do watch more. Each extra month of tenure is associated with about 1.4 more watch hours on average, and tenure explains roughly 71% of the variation in watch hours. The remaining 29% is likely down to plan type and region, which would be worth a follow-up."

One sentence with the finding, one sentence with the slope number in plain English, one sentence with R², one sentence naming what's not accounted for. That's a finding ready for a dashboard.

SHOW THIS OFF

Read the equation as a *sentence*, not as algebra. *"For every extra month of tenure, watch hours go up by 1.4 on average, and the line as a whole explains 71% of what's going on."* The assessor doesn't want the formula recited; they want to hear what it *means*.



SHARPEN YOUR PENCIL interpret these — no Excel needed

For each result, write a one-sentence interpretation in plain English. Imagine Priya is reading over your shoulder.

1. Avg session length vs avg user rating per title: $y = 0.4x + 3.1$, $R^2 = 0.08$. What does this tell you?

2. Tenure (months) vs total watch hours per subscriber: $y = 1.4x + 9.2$, $R^2 = 0.71$. What's the slope mean in plain English?

3. Number of devices used vs total watch hours per subscriber: $y = 14.2x - 5.3$, $R^2 = 0.46$. Strong, moderate, or weak — and what's your caveat?

Answers — check yourself

1. Essentially nothing. $R^2 = 0.08$ means session length tells you almost nothing about how a title is rated. The equation exists but the line isn't trustworthy.
2. Every extra month of tenure is associated with about 1.4 more watch hours on average. (Note "associated with", not "causes".)
3. Moderate. $R^2 = 0.46$ — more devices does tend to go with more hours, but it explains less than half the variation. Caveat: heavier viewers probably use more devices *because* they watch more, not the other way round. Lurking variable: engagement level itself.

There are no *Dumb* Questions

Q: *My scatter plot is curved, not straight. Can I still fit a linear trendline?*

A: You can, but the R^2 will be low and the line will be a poor summary. Excel's Add Trendline pane offers other options (exponential, polynomial, power, logarithmic). If the data clearly curves, pick the matching shape. If you're not sure which to pick, the eye-test is reasonable — fit two or three options, see which R^2 is highest, but be cautious: a higher-order polynomial will always fit more closely. Don't over-fit.

Q: *What if Y depends on more than one X? Can Excel handle that?*

A: That's multiple regression — beyond Level 3 scope, and beyond what a trendline can show. The honest answer when you suspect multiple drivers: name the other likely variables in your caveat ("*plan type and region likely matter too*"), and either segment your scatter plot (one per plan type) or flag it as a question for further analysis.

Q: *What's a "good" R^2 ?*

A: Depends on the field. Physics often sees 0.95+; behavioural data (how people watch TV) might top out at 0.4–0.6 and still be valuable. The number isn't good or bad on its own — what matters is whether it's useful for the decision.

Q: *Why does SLOPE want Y first, then X?*

A: Annoying, isn't it? Excel uses *known_y's* and *known_x's* — known outputs first, known inputs second. Just memorise that all three functions (SLOPE, INTERCEPT, RSQ) take Y first. If you swap them you'll get nonsense — typically a wildly different slope and a similarly wrong intercept.

Q: *Should regression go on the dashboard, or in supporting notes?*

A: The chart with the trendline can go on the dashboard if it tells a clear story. The bare equation usually doesn't — it belongs in supporting documentation. Show the picture, narrate the finding ("*longer-tenure subscribers watch more*"), and have the equation and R^2 available if anyone asks.

Three numbers — slope, intercept, R^2 . One sentence in plain English. One caveat. That's a finding.

Say it out loud

Regression is one of those topics where the assessor wants to hear plain English, not algebra. Practise these three out loud.

Q1 — Walk me through what your trendline equation means.

Read it as a sentence, not as maths. *"The equation is $y = 1.4x + 9.2$. The slope of 1.4 means each extra month of tenure is associated with about 1.4 more watch hours. The intercept of 9.2 means a brand-new subscriber is predicted to watch about 9.2 hours."* Always translate slope and intercept into the units of your actual columns.

Q2 — How trustworthy is your line?

This is the R^2 question, in disguise. Two sentences. *" R^2 is 0.71, which means tenure explains about 71% of the variation in watch hours. The remaining 29% is down to other factors not in this analysis — plan type and region being two likely candidates."* Always name what the analysis doesn't account for.

Q3 — What could be a lurking variable?

The shape: name a plausible third factor and explain why it might matter. For tenure vs hours: *"Plan type. Premium subscribers might both stay longer (higher tenure) AND watch more (more hours) because they're more engaged in the first place. Tenure and hours would correlate, but the underlying driver is engagement, not tenure itself."* Naming lurking variables is what makes a competent analysis look like a senior one.



DISTINCTION TELL

For every regression you present, narrate **the three numbers + one caveat**. Slope (what each unit of X buys you). Intercept (the starting value). R^2 (how trustworthy the line is). Caveat (what's not in the analysis). Four sentences. That's distinction band.

Bullet Points — the chapter in 30 seconds

- **Scatter plot first.** Two columns, one chart, eye-test before maths.
- **Line of best fit summarises the pattern in two numbers** — slope (m) and intercept (c).
- **R² is the trust score.** 0 = no pattern, 0.7+ = strong, 1.0 = perfect (and probably suspicious).
- **Three formulas:** `=SLOPE` , `=INTERCEPT` , `=RSQ` . Y comes first. Always.
- **Correlation isn't causation.** Every regression finding ends with a caveat about lurking variables.
- **Translate into plain English.** Read the equation as a sentence, not as maths.

TEST DRIVE

Before closing the laptop, run this self-check. Tick each box only when it's true:

- ☐ You built at least one scatter plot from a pair of columns in the StreamWave data.
- ☐ You added a trendline with equation and R² displayed on the chart.
- ☐ You can interpret your equation in one sentence of plain English.
- ☐ You can say what your R² value means — and whether you'd lean on the line.
- ☐ You named at least one plausible lurking variable for the relationship you analysed.
- ☐ You used `=SLOPE` , `=INTERCEPT` , and `=RSQ` at least once each in a summary table.
- ☐ You ran the **Two-Route Check** on the regression — eyeball, predict-and-compare, and chart-vs-formula numbers match.

COMING UP — DAY 6

Pivots and Charts. Tomorrow we turn the week's analysis into something visual. PivotTables for slicing the data ten ways without writing ten formulas. PivotCharts for the headline visuals. *The Chart Test* — choosing the right chart for the question being asked. Refresh — one click to rerun the whole pipeline. The bridge between today's regression numbers and Friday's polished dashboard.

that's Day 5 done — the data has revealed a pattern.